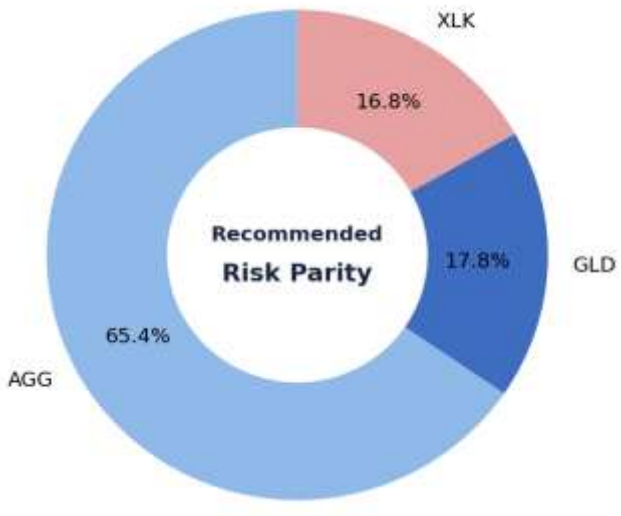
SERENDIB UNIVERSITY ENDOWMENT FUND

Recommended strategic model portfolio for a USD 10 million conservative institutional endowment

RECOMMENDATION: Adopt the Best Risk Parity portfolio - 65.45% AGG, 17.79% GLD and 16.76% XLK. It best balances positive return potential, validated downside-risk measurement, diversification and stress resilience.



Recommended Allocation

Asset	Weight	Portfolio role
AGG	**65.45%**	Defensive fixed-income anchor
GLD	**17.79%**	Real-asset / crisis diversifier
XLK	**16.76%**	Growth and upside participation

OOS return **6.51%** Annualized, 2022-2026	**OOS volatility** **8.19%** Annualized	**Sharpe ratio** **0.30** Positive risk-adjusted return	**Max drawdown** **-17.11%** OOS period
99% Hist. VaR **0.94%** ~USD 93.9k one-day loss cutoff	**99% Hist. ES** **1.46%** ~USD 145.8k avg. beyond VaR	**Stress rank** **#2 / 5** Combined stress framework	**Basel zone** **Green** 99% Historical VaR

WHY THIS PORTFOLIO

Risk Parity kept a positive OOS Sharpe ratio, ranked second under stress and avoided the >92% AGG concentration of GMVP and Minimum ES. Risk contributions are also much more balanced than capital weights.

HOW BAD COULD IT GET?

Historical replay: worst cumulative stress result was about -12.97% in the 2022 rate/inflation shock. The worst designed hypothetical shock was about -10.12%. These are scenario losses, not forecasts.

HOW CONFIDENT ARE WE?

Historical VaR passed Kupiec, independence and conditional-coverage tests at 99%. 15 exceptions occurred in 1,203 forecasts (1.25% observed vs 1.00% expected). Passing does not guarantee future accuracy.

Investment Committee Action

Adopt as the strategic model allocation, subject to periodic governance review. Review or rebalance if risk-adjusted performance deteriorates persistently, VaR validation fails repeatedly, concentration rises materially, or stress/deep-tail estimates increase sharply.

Source: Quant_Colombo Notebooks 01-09 and final investment memo. OOS evaluation begins 3 Jan 2022. Risk measures shown are one-day unless stated otherwise.
Important: stress tests are deterministic/historical scenario analyses; model outputs are not guarantees of future performance.